

Lecture 3 — The Execution Model: Stack, Heap, and the Process Address Space

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This lecture is not yet written. The outline lives in the plan file.

Intended sections:

- 1 The process address space
- 2 Anatomy of a stack frame
- 3 Calling conventions (System V AMD64 ABI)
- 4 How `execve` builds the initial stack
- 5 Per-thread stacks and guard pages
- 6 Two stacks per task — user and kernel
- 7 Per-thread non-stack state: signals and TLS
- 8 Security mechanisms on the stack (forward to L05)